THE LAWS OF SOFTWARE TEAMS

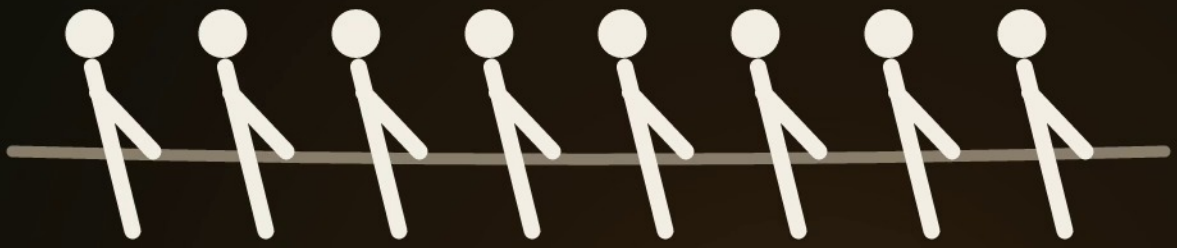
The Ringelmann effect

Individual effort drops as a group gets bigger

swipe →

Ivan Rykov

8 people pulled a rope
They pulled like 4



total pull, in people

of 8

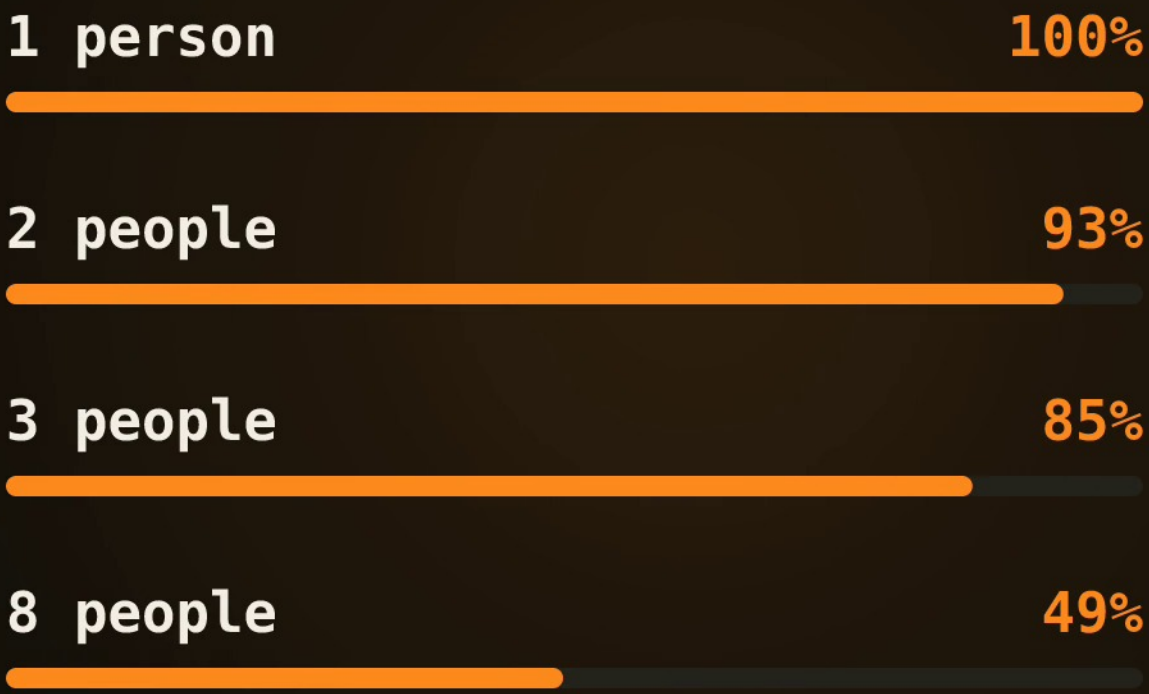
3.9

8 × 49% ≈ 3.9 · Ringelmann, 1913

Max Ringelmann

French agricultural engineer
Tested in the 1880s, published 1913

effort per person, by group size



Two things go wrong

1

Coordination

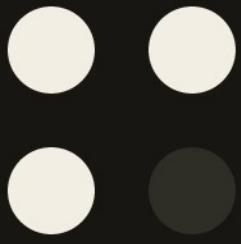
Nobody pulls at the same moment



2

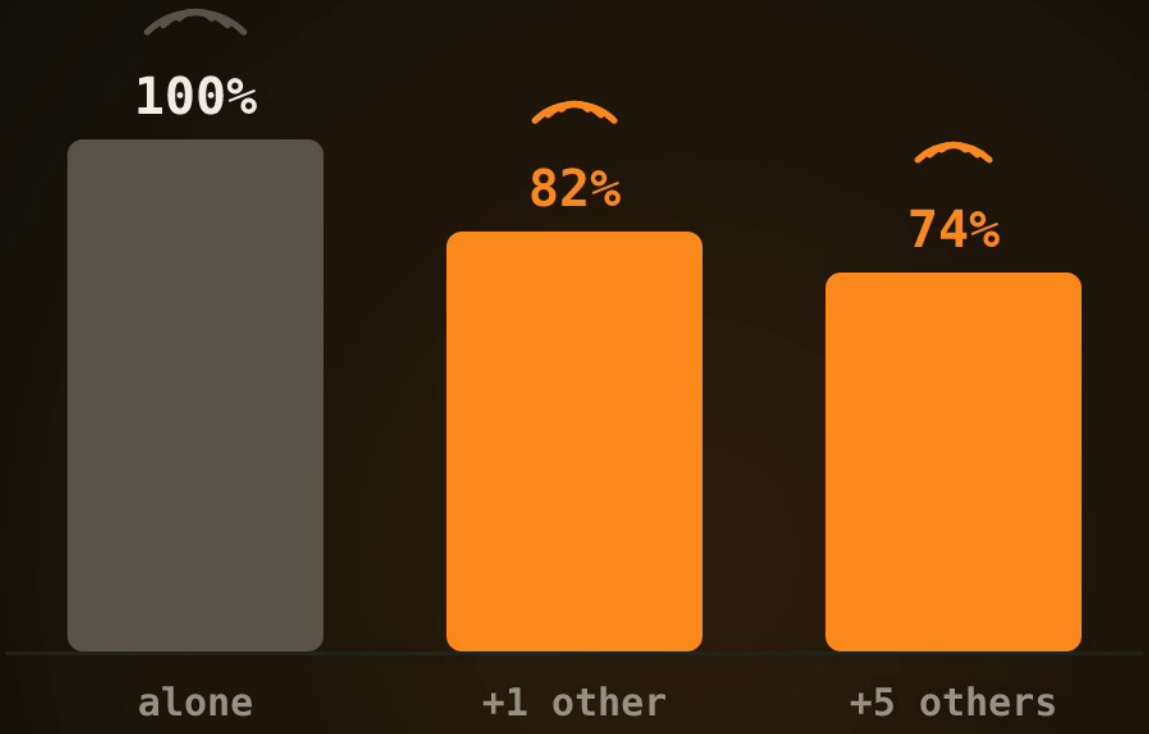
Motivation

“Someone else will pull”



1979: people shouted alone, blindfolded

But told others were shouting too:

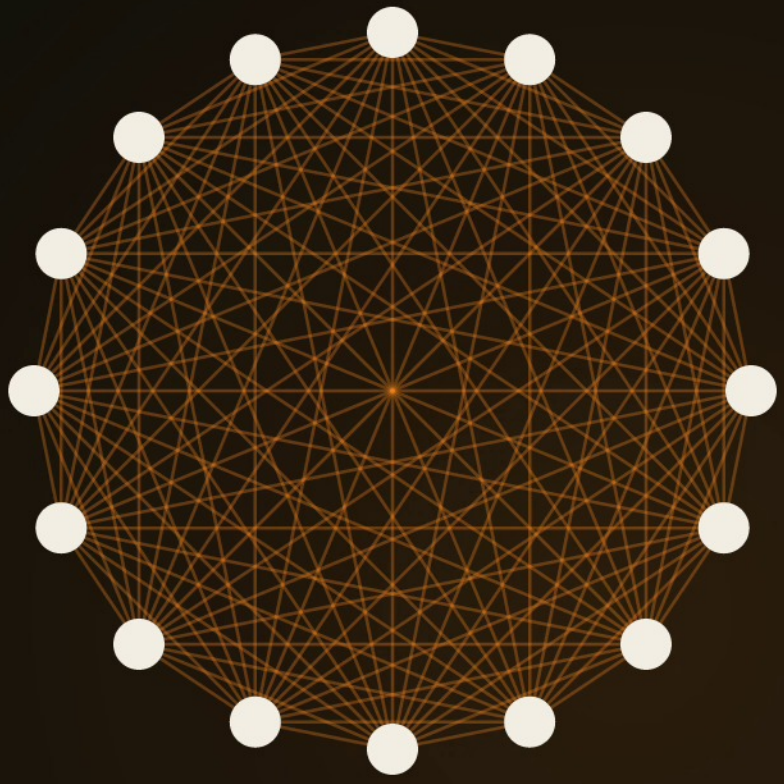


Nobody to coordinate with
They just tried less

Latané, Williams & Harkins, 1979 · "social loafing"

05 IN SOFTWARE

In software, everyone talks to everyone



16 people → 120 lines

n(n−1)/2 · Brooks, The Mythical Man-Month, 1975

Ivan Rykov

The data agrees

390 software projects, same size:
teams of 9+ vs teams under 4

−30% schedule

+350% cost

+500% defects

Best results: teams of 3-7

491 projects, Putnam 1997

Doug Putnam, QSM

07 THE FIX

How to beat it:

1. **Keep teams small**
   3-7 people, clear ownership

2. **Make each part visible**
   loafing hides in the crowd

3. **Give work that matters**
   effort follows meaning

Karau & Williams, 1993 · review of 78 studies

PostHog: 47 people, 15 teams

via manager.dev

So what's the right size? Small enough that nobody can hide



TAKEAWAY

So what's the right size? Small enough that nobody can hide

Sources: Kravitz and Martin, 1986 · Latané, Williams and Harkins, 1979 · QSM, Putnam, 1997

next: Price's law